Assignment 9A. Lagrange's interpolation and least square Marks 20

Posted on 05.11.2026 @ 2:30 pm and due on 05.11.2025 @ 5:40 pm

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1. Code library function for Lagrange's interpolation formula, which accepts order of polynomial and dataset as arrays / matrices. **[3]**

2. Write a library function for linear regression, which will also calculate Pearson's correlation coefficient r^2. **[4]**

3. Use Lagrange's interpolation formula to find the best estimate for $y(x = 6.7)$ from the data table below. **[5]**

x	2	3	5	8	12
$y(x)$	10	15	25	40	60

4. Fit the data given in the table below with power law $y = ax^b$ and exponential $y = ae^{-bx}$ models. Based on Pearson's r^2, determine which model gives better fit. Show the data points along with the fits in a plot. **[8]**

x	2.5	3.5	5.0	6.0	7.5	10.0	12.5	15.0	17.5	20.5
y	13.0	11.0	8.5	8.2	7.0	6.2	5.2	4.8	4.6	4.3